**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**

ACADEMIC ASSIGNMENT

# CO5 - AT 1

# CONSTRAINT-BASED PROBLEM SOLVING

*Network Performance: Constraint-Based Problem Solving*

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**Declaration**

*I hereby declare that this assignment is my own original work, prepared under the guidance of my course faculty, and has been submitted in accordance with the academic requirements of the course.*

## Question 1: Rural Branch Office - Reliable and Secure Internet Connectivity

A multinational company opening a branch in a remote rural area has four major constraints: high-speed wired internet is unavailable, the office must run cloud applications and video conferences, communication with headquarters must remain secure, and the connectivity budget is limited. The network should therefore use a solution that gives acceptable bandwidth, business continuity, reasonable cost, and room for future expansion.

### Recommended Internet Connection and Devices

- Use a primary 4G/5G Fixed Wireless Access connection where cellular coverage is stable. It is faster to deploy than laying new fiber and generally costs less than a dedicated satellite-only solution.
- Use satellite internet as the backup link when cellular service is unavailable or unstable. A dual-WAN business router can automatically switch traffic from the primary link to the backup link during failure.
- Deploy a managed Gigabit switch for office systems, Wi-Fi 6 access points for laptops and mobile devices, and a UPS for the modem, router, switch, and access point so short power failures do not interrupt work.

### Security Mechanisms

- Configure a site-to-site VPN between the branch and headquarters so sensitive business traffic is encrypted while travelling over public internet links.
- Enable a next-generation firewall on the edge router to control inbound and outbound traffic, block unauthorized connections, and separate business traffic from guest Wi-Fi.
- Use WPA3/WPA2-Enterprise for wireless access and strong user authentication. Keep network devices patched and restrict administrative access to authorized staff only.

### Bandwidth and Performance Management

Because rural links may have limited capacity, Quality of Service (QoS) should prioritize video conferencing, VoIP, VPN traffic, and cloud applications over software downloads and non-business browsing. Large updates and backups can be scheduled outside working hours. Monitoring bandwidth utilization will help the administrator identify congestion and adjust policies before users experience poor performance.

### Justification

This design balances all stated constraints. Fixed wireless keeps the primary connection affordable, while satellite backup improves reliability and supports uninterrupted operations. The dual-WAN router provides automatic failover, the VPN and firewall protect communication with headquarters, and QoS makes better use of limited bandwidth. The design is also scalable: when fiber or another high-speed service becomes available later, it can replace the primary WAN link without changing the internal network.

## Question 2: Campus-Wide Wireless Network for More Than 3,000 Students

The university must provide wireless access to more than 3,000 students in classrooms, laboratories, libraries, and hostels. The main constraints are a very large number of simultaneous users, limited access points, interference between wireless cells, secure authentication, and a restricted budget. The design must therefore maximize coverage and capacity from every access point instead of simply increasing the number of devices.

### Access Point Placement

- Carry out a wireless site survey and place access points at central, elevated locations where users are concentrated. High-density areas such as libraries, large classrooms, laboratories, and hostel common areas should receive priority.
- Avoid placing access points behind thick walls, metal objects, electrical equipment, or other major obstacles. Adjacent access points should have partially overlapping coverage so users can roam without losing connectivity.
- Use controller-based or centrally managed access points so transmit power and channel assignments can be adjusted from one location. This reduces interference and makes efficient use of the limited AP count.

### Frequency Band and Channel Planning

- Prefer the 5 GHz band for high-density zones because it provides more non-overlapping channels and usually experiences less interference than 2.4 GHz. It is suitable for classrooms, labs, and libraries where many users access the network simultaneously.
- Retain 2.4 GHz mainly for wider coverage and older devices because it travels farther through walls, but use non-overlapping channels such as 1, 6, and 11 to minimize co-channel interference.
- Where supported, Wi-Fi 6/6E features can improve efficiency by serving many users more effectively, but deployment should remain consistent with the available budget and client-device support.

### Authentication and Security

Use WPA2-Enterprise or WPA3-Enterprise with 802.1X authentication and a central RADIUS server. Students should authenticate with university credentials instead of sharing a common Wi-Fi password. Separate student, staff, laboratory, and guest traffic using VLANs and apply access-control rules so users can reach only the services they are authorized to use.

### Balancing Coverage, Performance, Security, and Cost

The solution reduces cost by placing fewer access points strategically instead of installing many APs without planning. The 5 GHz band provides capacity in crowded zones, while 2.4 GHz fills larger coverage areas.

## Question 3: Secure Network Architecture for a Financial Institution

A financial institution must protect sensitive customer information and online banking services while maintaining good network performance and controlling operational cost. The design must therefore satisfy four constraints at the same time: strong security, regulatory compliance, low performance overhead, and affordable implementation. A layered security architecture is appropriate because no single control can protect every part of the network.

### Network Segmentation

- Create separate VLANs for departments such as Finance, Customer Service, Administration, IT, servers, and guest devices. Sensitive banking databases should be placed in a protected server VLAN or data-center segment.
- Control communication between VLANs using Layer 3 access-control lists or firewall policies. Only required traffic should be allowed between departments, reducing both unnecessary broadcasts and the possibility of lateral movement after a security incident.
- Place public-facing services in a DMZ so internet users do not directly access the internal banking network.

### Firewall Deployment

Deploy an enterprise firewall at the internet edge between the internal network and external networks. The firewall should use a default-deny approach, allowing only authorized services. A second internal firewall or segmented firewall zones can protect critical servers without requiring a separate physical firewall for every department, which helps control cost. Stateful inspection, application control, URL filtering, and secure logging improve protection while preserving manageable performance.

### Authentication and Access Control

Employees accessing sensitive customer data should use unique user accounts, role-based access control, and multi-factor authentication. Privileged administrator accounts must be separated from normal user accounts. Strong password policies and centralized identity management reduce unauthorized access and help the institution maintain an auditable record of who accessed important systems.

### Intrusion Detection and Monitoring

Use an IDS/IPS at key points such as the internet edge and the server network. The system should inspect suspicious traffic and generate alerts for attacks or abnormal behaviour. Firewall, VPN, authentication, and IDS/IPS logs should be collected centrally so administrators can review incidents and maintain evidence required for security audits and compliance.

**Justification**

This architecture improves security without unnecessarily reducing performance. VLAN segmentation limits broadcast traffic and isolates sensitive systems, while firewall rules restrict access only where necessary. Multi-factor authentication strengthens user security, and IDS/IPS provides continuous threat detection. Using centralized management and logical segmentation reduces the need for excessive hardware, helping control operational cost. The architecture can also scale by adding new VLANs, firewall zones, or server segments as the institution grows. Therefore, the proposed design provides a feasible balance of security, performance, compliance, and budget while protecting sensitive financial data and online banking services.